

Title

Cargo Tank Venting & Inert Gas System & Equipment

Title

Inert Gas System Introduction

Image

- ☐ *Nature of tanker cargoes*
- ☐ *Hydro-carbon and Air mixtures*
- ☐ *May result in fire & explosion*
- ☐ *Efforts on tankers are towards:*
 - (a) Eliminate sources of Ign.*
 - (b) Eliminate flamm. periods*
- ☐ *Objective (a) acheived but*
- ☐ *Objective (b) possible*

Provision of an IGS can give absolute control

Title

Introduction to Inert Gas System

Page-text

➤ Introduction

Page-text

➤ Flammability

Page-text

➤ Flammable Range

Page-text

➤ Effect of Increasing Oxygen

Page-text

➤ Suppression of Flammability by IG

Page-text

➤ Combustion & Explosion

Page-text

➤ Inert Gas Prevents FIRE & EXPLOSION

ADVANTAGES OF INERT GAS

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- MAINTAINS NON FLAMMABLE ATMOSPHERE IN THE CARGO TANKS.

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- BY MAINTAINING A POSITIVE PRESSURE IN THE TANKS WITH I.G. IT WILL ASSIST IN EXPEDITING DISCHARGE OF CARGO.

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- REDUCES CORROSION LEVEL IN THE TANKS DUE TO OXYGEN DEFICIENCY HENCE REDUCED OXIDATION.

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- ASSISTS DURING GAS FREEING TO AVOID PASSING THRU THE FLAMMABLE REGION.

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MAIN DISADVANTAGE

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- DUE TO LOW O_2 CONTENT IN INERTED TANKS I.G. POSES A GRAVE HEALTH HAZARD.

PORTABLE GAS DETECTION INSTRUMENTS

OXYGEN ANALYSER

Measures OXYGEN by % volume.
(Fresh air contains 21% oxygen)

PERSONAL OXYGEN ALARM

Normally set to alarm if OXYGEN level falls to 18% volume.

**COMBUSTIBLE GAS
INDICATOR (Explosimeter)**

Measures HYDROCARBON GAS as percentage of Lower Flammable Limit in air. (Will NOT measure Hydrocarbon gas in an Inert Atmosphere)

TANKSCOPE

Measures ANY HYDROCARBON GAS up to 20% volume in inert gas or air.

GASCOPE

Measures SPECIFIC HYDROCARBON GASES up to 100% volume in nitrogen

**MULTIGAS DETECTOR
GASTEC ETC.**

With appropriate tubes can be used to measure a wide range of GASES either as % volume or in parts per million.
(Consult tube specification)

NOTES

1. OTHER INSTRUMENTS MAY BE USED FOR MEASURING SPECIFIC GASES.
2. ALL THE ABOVE INSTRUMENTS NEED CHECKING AND MOST NEED CALIBRATION BEFORE USE.
3. PERSONAL OXYGEN ALARMS ARE NOT DESIGNED TO BE USED FOR CHECKING LOW OXYGEN LEVELS IN INERT GAS.
4. WHERE COMBINATION OF EXPLOSIMETERS ARE PROVIDED, THEY SHOULD BE USED FOR CHECKING SPACES WHICH ARE THOUGHT TO BE GAS FREE, PRIOR TO ENTRY.

MSS**Marine Safety Services**

Title

Image

Suppression of Flammability by IG



*Suppression by (a) Flue gas
(b) Inert gas*



Narrowing of flammable range



Increase of min. ignition energy

If the oxygen availability is reduced to a sufficient extent, the mixture will become non-flammable no matter what the combustible vapour content may be.

Title

Suppression of Flammability by IG

Image

Suppression of Flammability



*Suppression by (a) Flue gas
(b) Inert gas*



Narrowing of flammable range



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Effect of Oxygen on Flammable Range

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Vapour	Limits in Air % by Volume		Limits in Oxygen % by Volume	
	Lower	Upper	Lower	Upper
<i>Propane</i>	2.2	9.5	2.3	55.0
<i>Butane</i>	1.9	8.5	1.8	49.0

The introduction of oxygen to a flammable vapour results in broadening the flammable range

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Inert Gas Definitions

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Inerting

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... means the introduction of inert gas into a tank with the object of attaining the inert condition (Oxygen content < 8% by vol.)



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Gas-freeing

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... means the introduction of fresh air into a tank with the object of removing toxic, flammable and inert gasses and increasing the Oxygen content to 21% by volume.



Obj-text

Purging

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... means the introduction of inert gas into a tank already in the inert condition with the object of further reducing the oxygen content reducing the existing Hcn gas content

Inert Gas Must On Tankers

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➤ Required Under Regulation 60 of SOLAS

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➤ For All Petroleum Tankers more than 20,000 DWT –
Keel laid after 1-9-84

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➤ Exception for Existing Tankers (Keel Laid prior 84)

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- i) Tank Washing Machine Capacity less than 60 Cubic
Meter/Hour &

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- ii) Unreasonable & Impracticable due to design determined
by Administration

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- iii) For Product Tankers between 20-40,000DWT tank
washing Machine Capacity less than 60 cubic meter/Hour

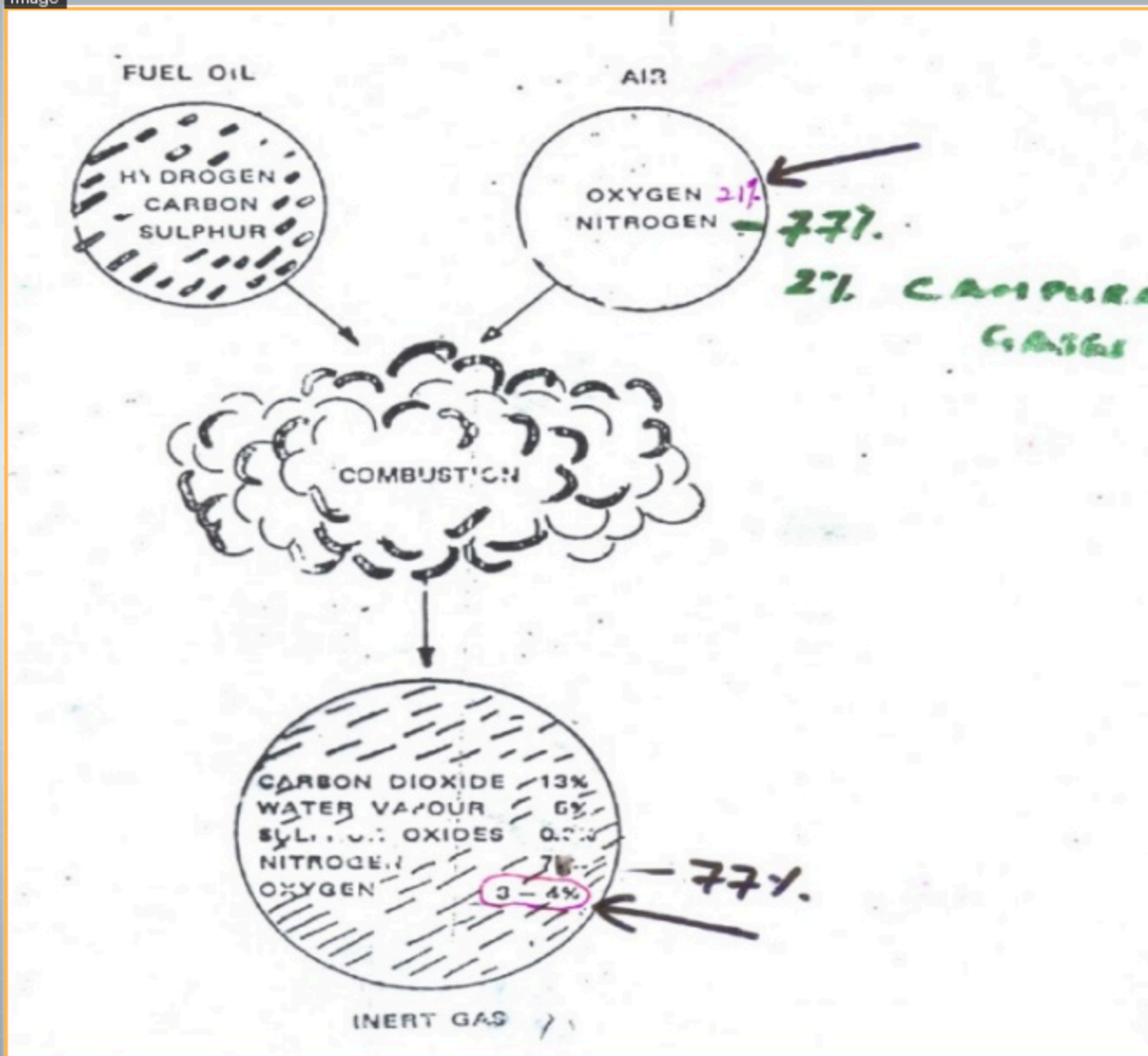
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➤ Tankers fitted with IGS to keep COT inerted at all times

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Inert Gas Production

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Title

Inert Gas System

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➤ **Flue Gas Uptake Valves**

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➤ **Scrubber**

Page-text

➤ **I.G. Fans (Blowers)**

Page-text

➤ **Gas Regulating Valve**

Page-text

➤ **Deck Water Seal**

Page-text

➤ **Non Return Valve**

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➤ **Deck Isolating Valve (Manual)**

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➤ **P/ V Breaker**

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➤ **Instrumentation**

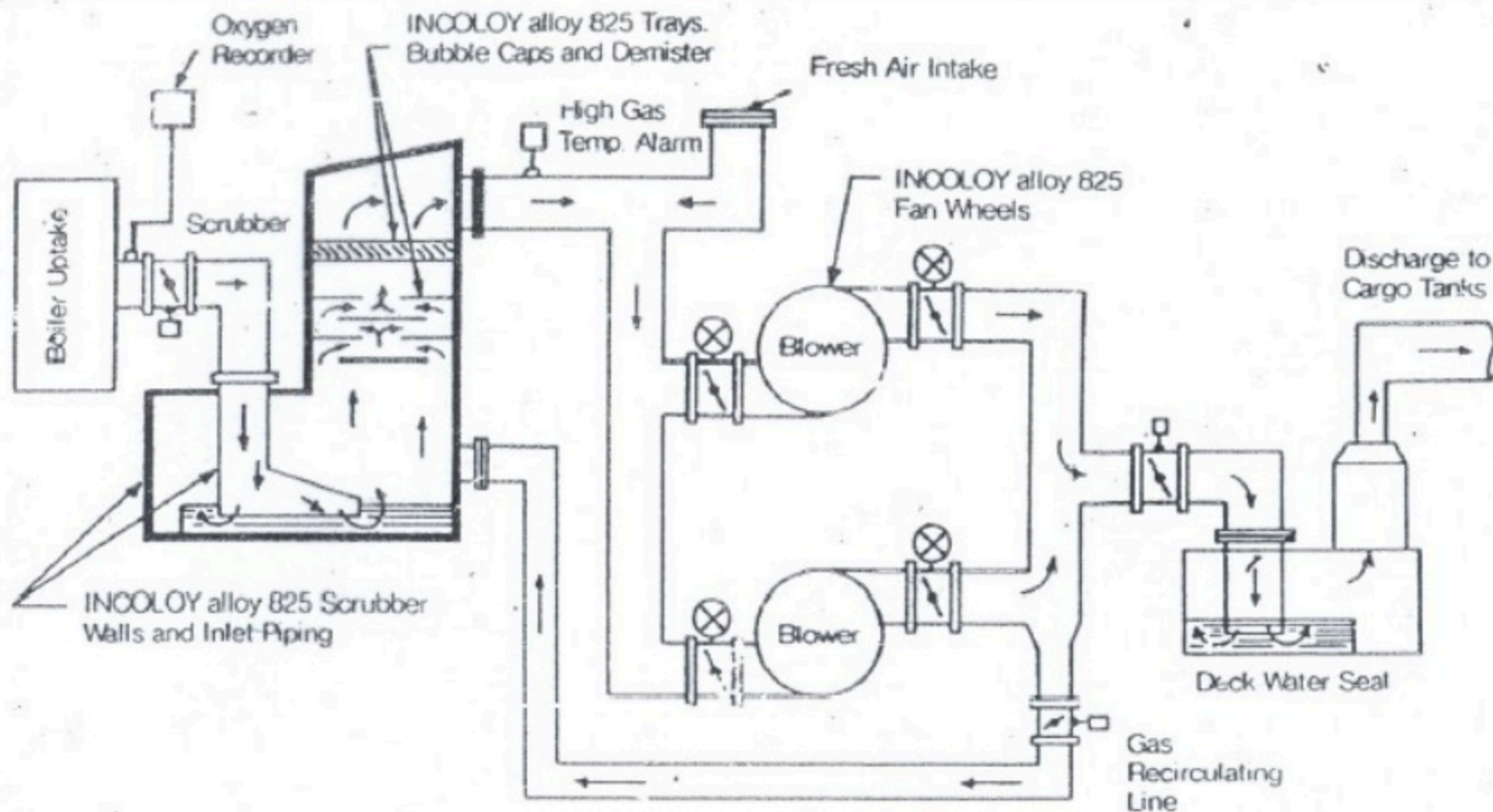
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➤ **Tank Atmosphere Analyzers (Oxygen Analyzers) Fixed & Portable**

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Inert Gas System Layout

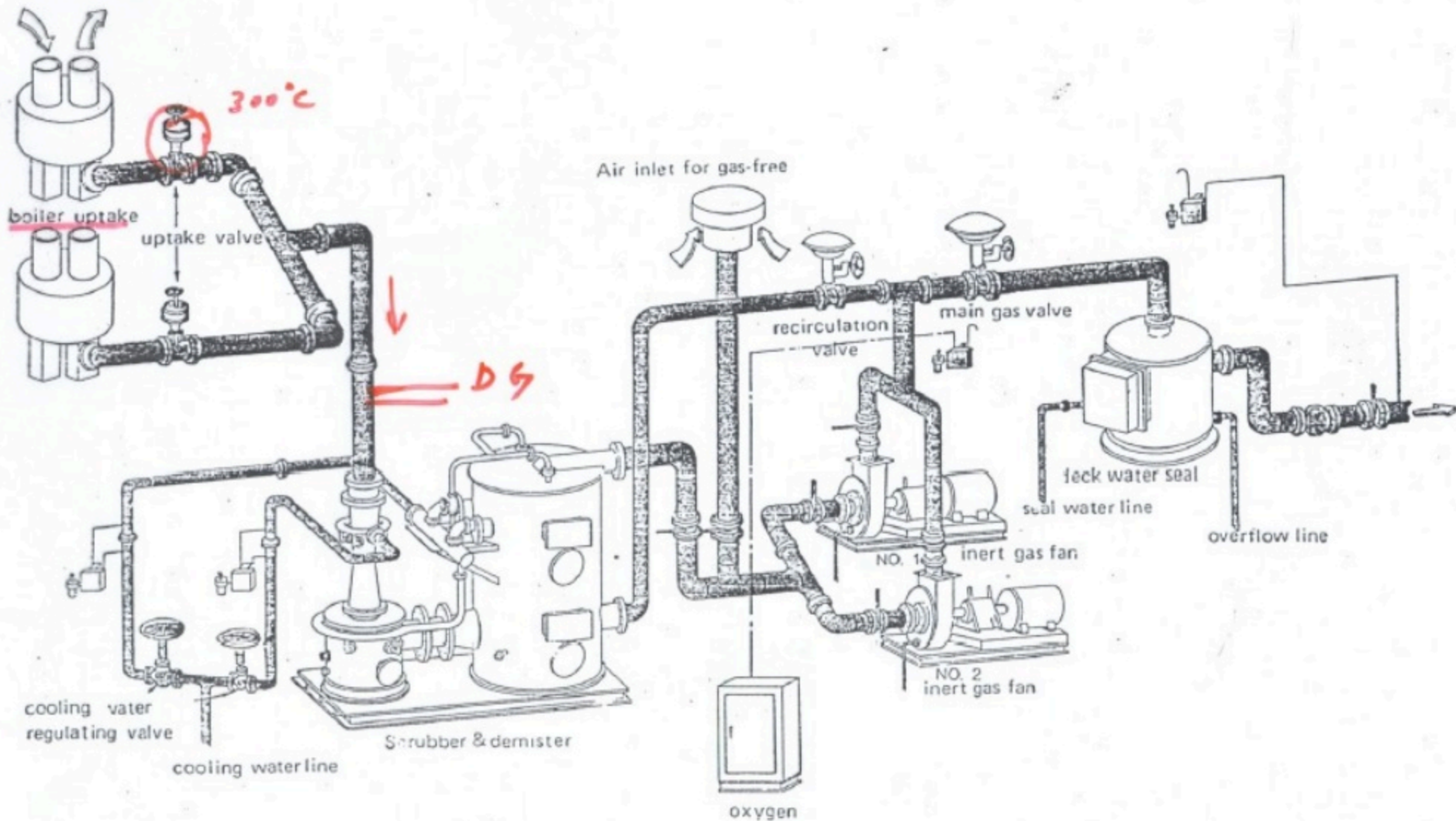
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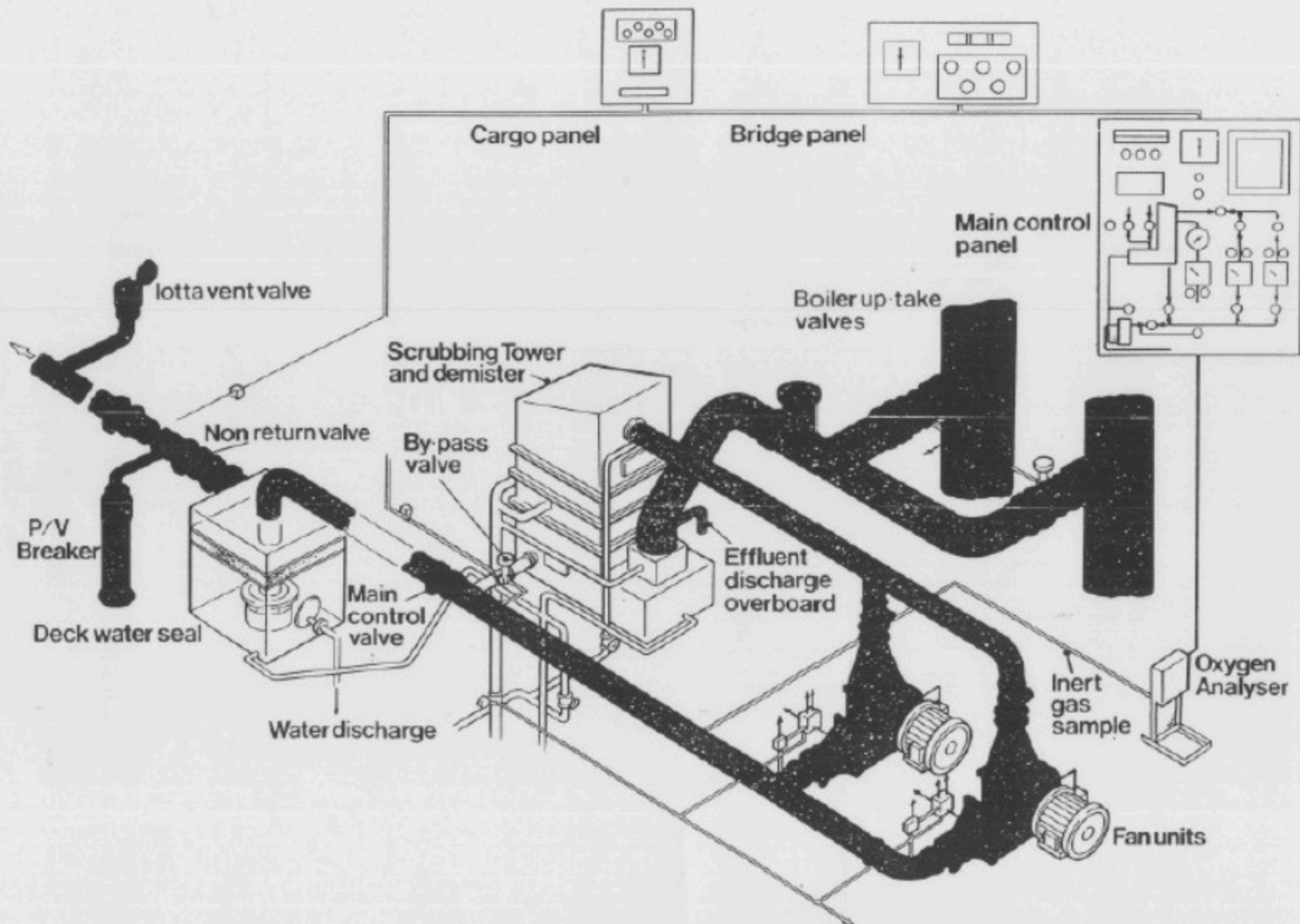
Inert Gas System Flow Sheet

Image



Inert Gas Lay Out

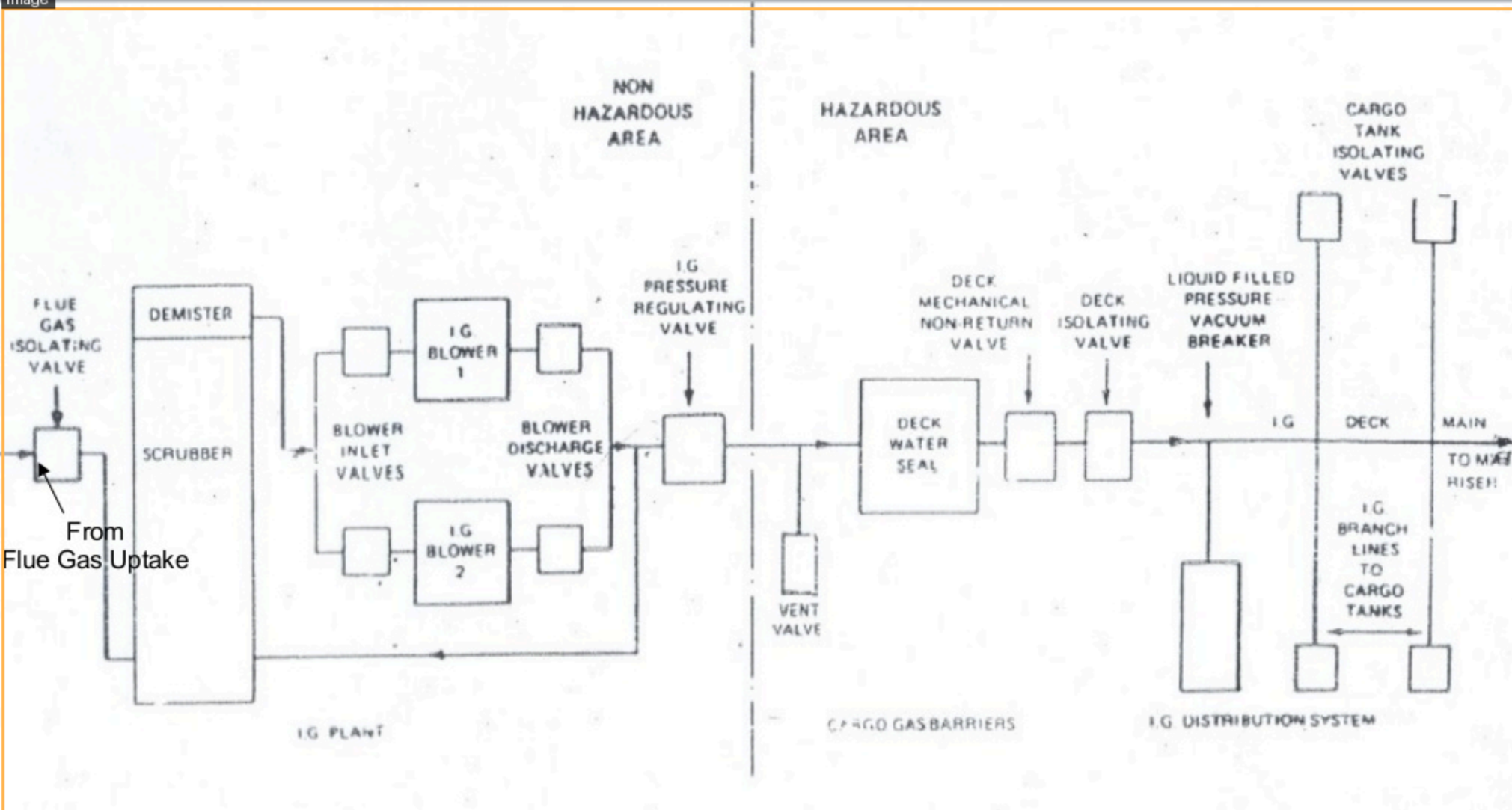
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Inert Gas Layout System

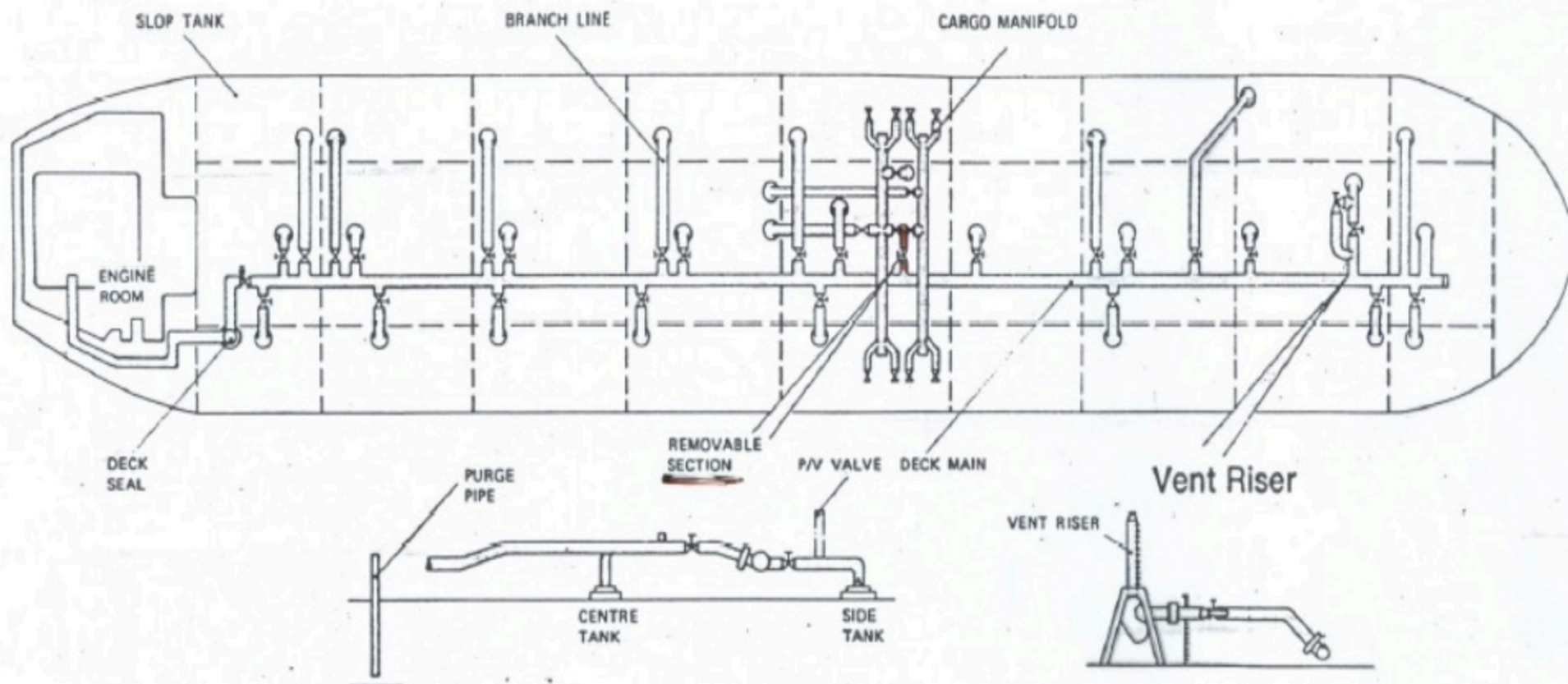
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Deck Distribution of Inert Gas System

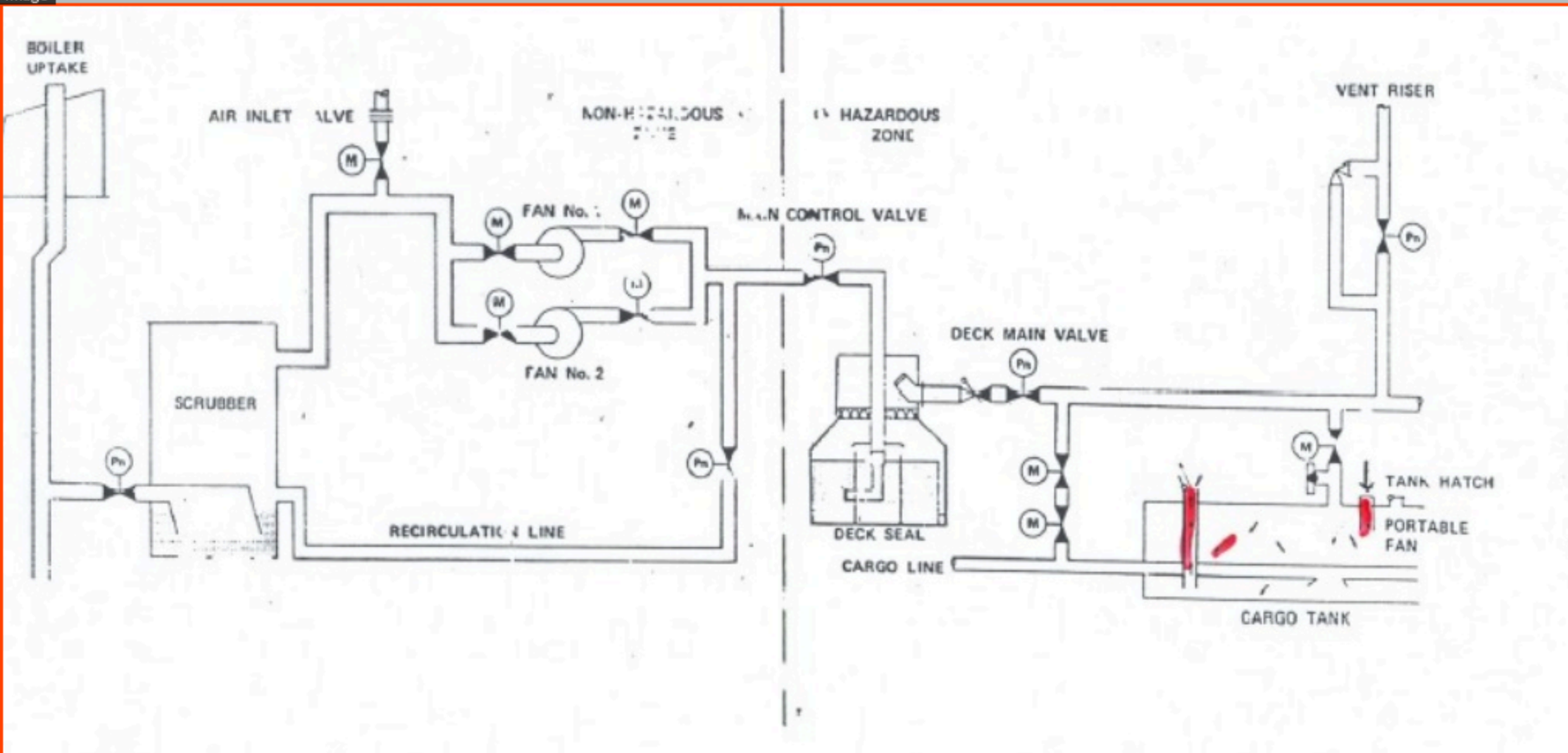
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I. G. S. LAYOUT

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Deck Water Seal Requirement

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- Regulation 62 requires that Inert Gas system is equipped with two Non return devices one of which shall be water seal to prevent the return of hydrocarbon to Gas safe spaces

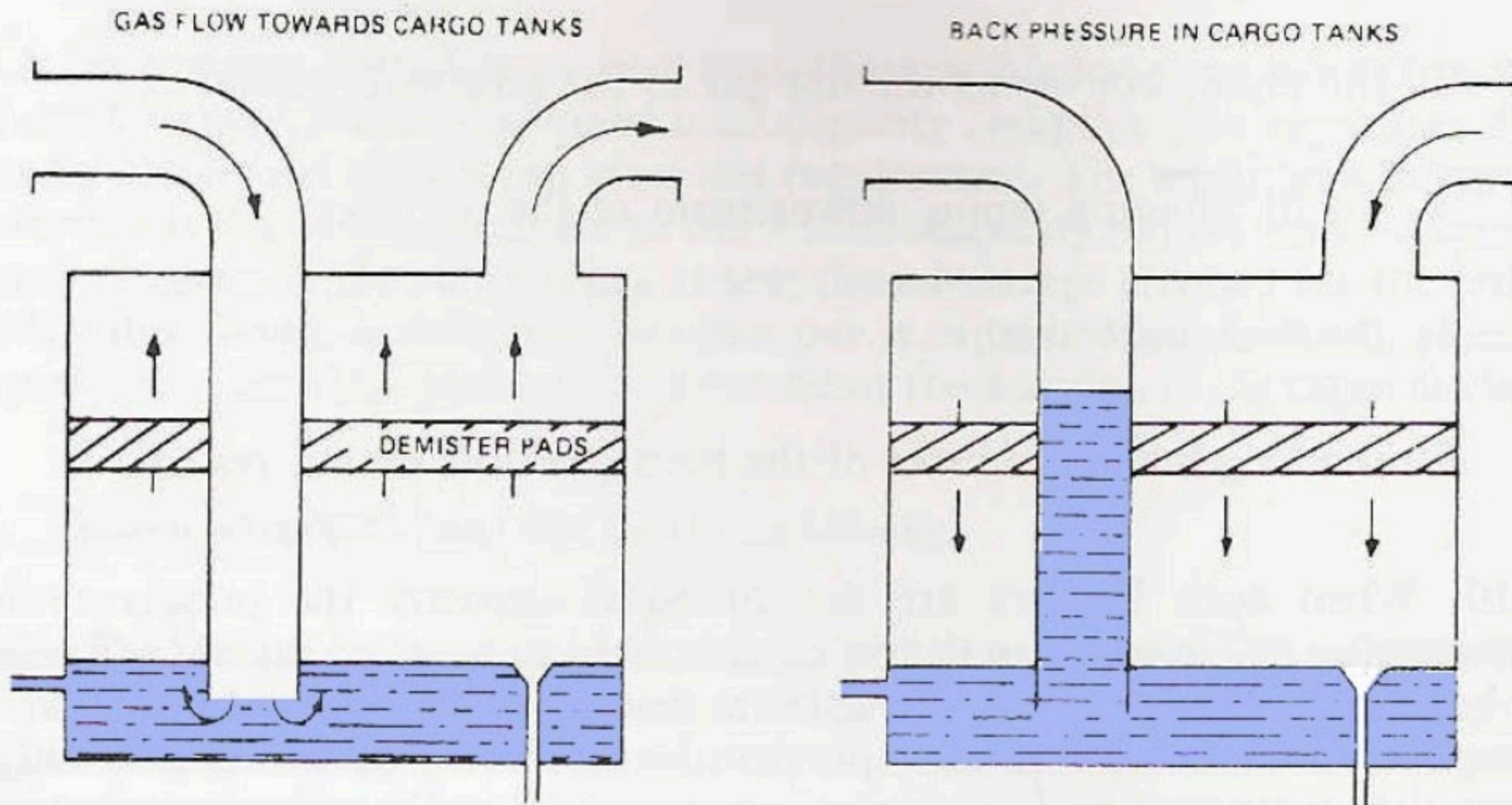
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- In addition Regulation 62 requires that arrangement of Deck Water Seal Shall be such that it will prevent back flow of hydrocarbon vapours and ensure proper Functioning of Seal Under Operating Conditions.

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Deck Water Seal Wet Type

Image



Deck water seal – wet type.